

Laplace Transform

$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

06016401

MATHEMATICS FOR INFORMATION TECHNOLOGY

Instructor: Asst. Prof. Dr. Praphan Pavarangkoon

Office: Room no. 418-5, 4th Floor

Email: praphan@it.kmitl.ac.th

Office hours: Wednesday at 13:00 – 16:00

or as an advance appointment

Outline



Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

- Matrices
 - The Inverse of a Matrix
- Determinants
 - The Determinant of a Matrix
 - Applications of Determinants
- Applications of Matrix Operations
 - Cryptography

The Inverse of a Matrix

A decorative image in the top right corner showing a blurred snippet of a document with the Laplace Transform formula: $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$.

An $n \times n$ matrix A is **invertible** (or **nonsingular**) if there exists an $n \times n$ matrix B such that

$$AB = BA = I_n$$

where I_n is the identity matrix of order n . The matrix B is called the (multiplicative) **inverse** of A . A matrix that does not have an inverse is called **noninvertible** (or **singular**).

Finding the Inverse of a Matrix

Find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix}.$$

Properties of Inverse Matrices



Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

If A is an invertible matrix, k is a positive integer, and c is a scalar not equal to zero, then A^{-1} , A^k , cA , and A^T are invertible and the following are true.

1. $(A^{-1})^{-1} = A$
2. $(A^k)^{-1} = A^{-1}A^{-1} \cdot \cdot \cdot A^{-1} = (A^{-1})^k$
3. $(cA)^{-1} = \frac{1}{c}A^{-1}, c \neq 0$
4. $(A^T)^{-1} = (A^{-1})^T$

If A and B are invertible matrices of size n , then AB is invertible and

$$(AB)^{-1} = B^{-1}A^{-1}.$$

The Determinant of a Matrix

- Every *square* matrix can be associated with a real number called its **determinant**.
- Historically, the use of determinants arose from the recognition of special patterns that occur in the solutions of systems of linear equations.

Definition of the Determinant of a 2×2 Matrix

The **determinant** of the matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

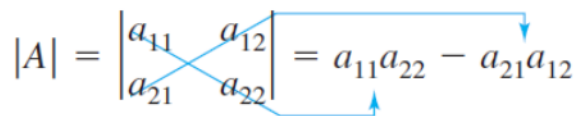
is given by

$$\det(A) = |A| = a_{11}a_{22} - a_{21}a_{12}.$$

REMARK: In this text, $\det(A)$ and $|A|$ are used interchangeably to represent the determinant of a matrix. Vertical bars are also used to denote the absolute value of a real number; the context will show which use is intended. Furthermore, it is common practice to delete the matrix brackets and write

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \quad \text{instead of} \quad \left| \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \right|.$$

A convenient method for remembering the formula for the determinant of a 2×2 matrix is shown in the diagram below.

$$|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12}$$


The determinant is the difference of the products of the two diagonals of the matrix. Note that the order is important, as demonstrated above.

The Determinant of a Matrix of Order 2

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$
with surface

Find the determinant of each matrix.

(a) $A = \begin{bmatrix} 2 & -3 \\ 1 & 2 \end{bmatrix}$ (b) $B = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$ (c) $C = \begin{bmatrix} 0 & 3 \\ 2 & 4 \end{bmatrix}$

Definitions of Minors and Cofactors of a Matrix

$$\text{Laplace-Trans} \\ f(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

If A is a square matrix, then the **minor** M_{ij} of the element a_{ij} is the determinant of the matrix obtained by deleting the i th row and j th column of A . The **cofactor** C_{ij} is given by

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

For example, if A is a 3×3 matrix, then the minors and cofactors of a_{21} and a_{22} are as shown in the diagram below.

Minor of a_{21}

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad M_{21} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$$

Delete row 2 and column 1.

Cofactor of a_{21}

$$\begin{aligned} C_{21} &= (-1)^{2+1} M_{21} \\ &= -M_{21} \end{aligned}$$

Minor of a_{22}

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad M_{22} = \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}$$

Delete row 2 and column 2.

Cofactor of a_{22}

$$\begin{aligned} C_{22} &= (-1)^{2+2} M_{22} \\ &= M_{22} \end{aligned}$$

Definitions of Minors and Cofactors of a Matrix (cont.)

As you can see, the minors and cofactors of a matrix can differ only in sign. To obtain the cofactors of a matrix, first find the minors and then apply the checkerboard pattern of +’s and -’s shown below.

Sign Pattern for Cofactors

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

3 × 3 matrix

$$\begin{bmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{bmatrix}$$

4 × 4 matrix

$$\begin{bmatrix} + & - & + & - & + & \dots \\ - & + & - & + & - & \dots \\ + & - & + & - & + & \dots \\ - & + & - & + & - & \dots \\ + & - & + & - & + & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

n × n matrix

Note that *odd* positions (where $i + j$ is odd) have negative signs, and even positions (where $i + j$ is even) have positive signs.

Find the Minors and Cofactors of a Matrix

A decorative image in the top right corner showing a blurred snippet of a document with the Laplace transform formula: $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$.
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

Find all the minors and cofactors of

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}.$$

Definition of the Determinant of a Matrix

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$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

If A is a square matrix (of order 2 or greater), then the determinant of A is the sum of the entries in the first row of A multiplied by their cofactors. That is,

$$\det(A) = |A| = \sum_{j=1}^n a_{1j}C_{1j} = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n}.$$

The Determinant of a Matrix of Order 3

A decorative image in the top right corner showing a blurred mathematical formula for the Laplace transform: $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$.
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

Find the determinant of

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}.$$

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Expansion by Cofactors



Laplace-Transform
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

Let A be a square matrix of order n . Then the determinant of A is given by

$$\det(A) = |A| = \sum_{j=1}^n a_{ij}C_{ij} = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$$

or

$$\det(A) = |A| = \sum_{i=1}^n a_{ij}C_{ij} = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}.$$

The Determinant of a Matrix of Order 4

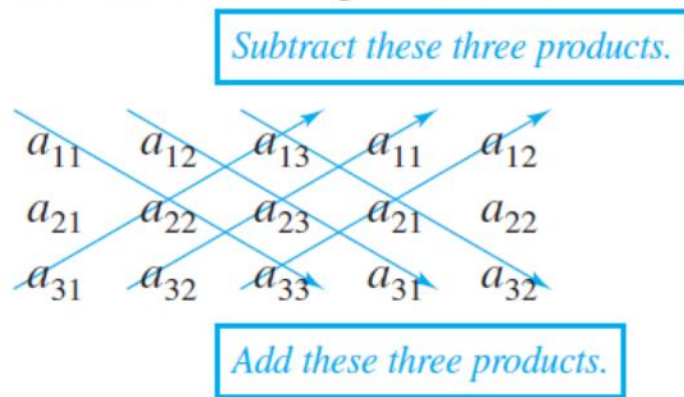
A decorative image in the top right corner showing a blurred snippet of a mathematical formula, likely the Laplace transform: $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$.
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

Find the determinant of

$$A = \begin{bmatrix} 1 & -2 & 3 & 0 \\ -1 & 1 & 0 & 2 \\ 0 & 2 & 0 & 3 \\ 3 & 4 & 0 & -2 \end{bmatrix}.$$

Definition of the Determinant of a Matrix (cont.)

There is an alternative method commonly used for evaluating the determinant of a 3×3 matrix A . To apply this method, copy the first and second columns of A to form fourth and fifth columns. The determinant of A is then obtained by adding (or subtracting) the products of the six diagonals, as shown in the following diagram.



Try confirming that the determinant of A is

$$|A| = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}.$$

The Determinant of a Matrix of Order 3

A decorative image in the top right corner showing a portion of a document with the Laplace-Transform formula: $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$.
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

Find the determinant of

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & -4 & 1 \end{bmatrix}.$$

Conditions That Yield a Zero Determinant

Laplace-Trans
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

If A is a square matrix and any one of the following conditions is true, then $\det(A) = 0$.

1. An entire row (or an entire column) consists of zeros.
2. Two rows (or columns) are equal.
3. One row (or column) is a multiple of another row (or column).

$$\begin{vmatrix} 0 & 0 & 0 \\ 2 & 4 & -5 \\ 3 & -5 & 2 \end{vmatrix} = 0,$$

*The first row
has all zeros.*

$$\begin{vmatrix} 1 & -2 & 4 \\ 0 & 1 & 2 \\ 1 & -2 & 4 \end{vmatrix} = 0,$$

*The first and third
rows are the same.*

$$\begin{vmatrix} 1 & 2 & -3 \\ 2 & -1 & -6 \\ -2 & 0 & 6 \end{vmatrix} = 0.$$

*The third column is a
multiple of the first column.*

Properties of Determinants

If A and B are square matrices of order n , then

$$\det(AB) = \det(A) \det(B).$$

If A is an $n \times n$ matrix and c is a scalar, then the determinant of cA is given by

$$\det(cA) = c^n \det(A).$$

A square matrix A is invertible (nonsingular) if and only if

$$\det(A) \neq 0.$$

If A is invertible, then

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

If A is a square matrix, then

$$\det(A) = \det(A^T).$$

The Adjoint of a Matrix



Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

- If A is a square matrix, then the **matrix of cofactors** of A has the form

$$\begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}.$$

- The transpose of this matrix is called the **adjoint** of A and is denoted by $\text{adj}(A)$. That is,

$$\text{adj}(A) = \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}.$$

Finding the Adjoint of a Square Matrix



Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

Find the adjoint of

$$A = \begin{bmatrix} -1 & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & -2 \end{bmatrix}.$$

The Inverse of a Matrix Given by Its Adjoint

A decorative image in the top right corner showing a snippet of a document with the Laplace transform formula: $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$.
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

If A is an $n \times n$ invertible matrix, then

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A).$$

Using the Adjoint of a Matrix to Find Its Inverse

Use the adjoint of

$$A = \begin{bmatrix} -1 & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & -2 \end{bmatrix}$$

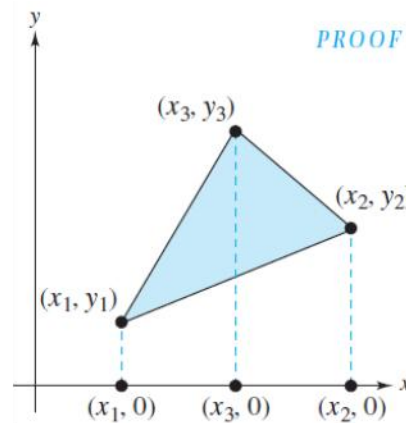
to find A^{-1} .

Applications of Determinants

- Area, Volume, and Equations of Lines and Planes
- Two-Point Form of the Equation of a Line
- Volume of a Tetrahedron

Area, Volume, and Equations of Lines and Planes

$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$



The area of the triangle whose vertices are (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) is given by

$$\text{Area} = \pm \frac{1}{2} \det \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix},$$

where the sign $(\pm)$ is chosen to yield a positive area.

Finding the Area of a Triangle

Find the area of the triangle whose vertices are $(1, 0)$, $(2, 2)$, and $(4, 3)$.

Two-Point Form of the Equation of a Line



Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

An equation of the line passing through the distinct points (x_1, y_1) and (x_2, y_2) is given by

$$\det \begin{bmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{bmatrix} = 0.$$

Finding an Equation of the Line Passing Through Two Points

Find an equation of the line passing through the points $(2, 4)$ and $(-1, 3)$.

Volume of a Tetrahedron

Laplace-Trans
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

- เตตระฮีดรอน ตรงกับภาษาอังกฤษว่า Tetrahedron (อ่านว่า tet-tra-he-dron) หมายถึงโครงสร้างปริมาตรฐานสามเหลี่ยม ซึ่งประกอบด้วย ด้าน (face) สามเหลี่ยม 4 ด้าน มุม (vertice) 4 มุม ขอบ (edge) 6 ขอบ เตตระฮีดรอนเป็นรูปทรงที่จัดอยู่ในประเภท พลาโตนิก (platonic)

The volume of the tetrahedron whose vertices are (x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) , and (x_4, y_4, z_4) is given by

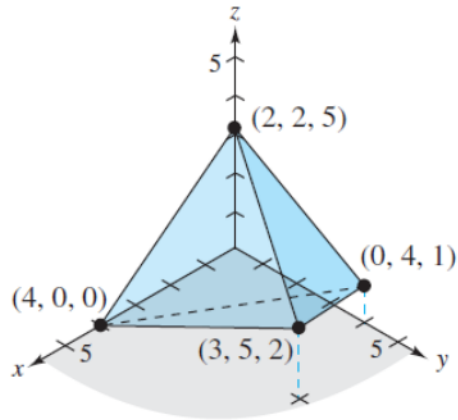
$$\text{Volume} = \pm \frac{1}{6} \det \begin{bmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{bmatrix},$$

where the sign $(\pm)$ is chosen to yield a positive volume.

Finding the Volume of a Tetrahedron

Laplace-Trans
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

Find the volume of the tetrahedron whose vertices are $(0, 4, 1)$, $(4, 0, 0)$, $(3, 5, 2)$, and $(2, 2, 5)$

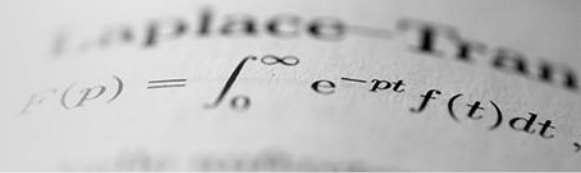


Applications of Matrix Operations

- Cryptography

A decorative header image showing a close-up of a document with the Laplace Transform formula. The text "Laplace-Trans" is visible at the top, followed by the equation $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$.
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

Cryptography

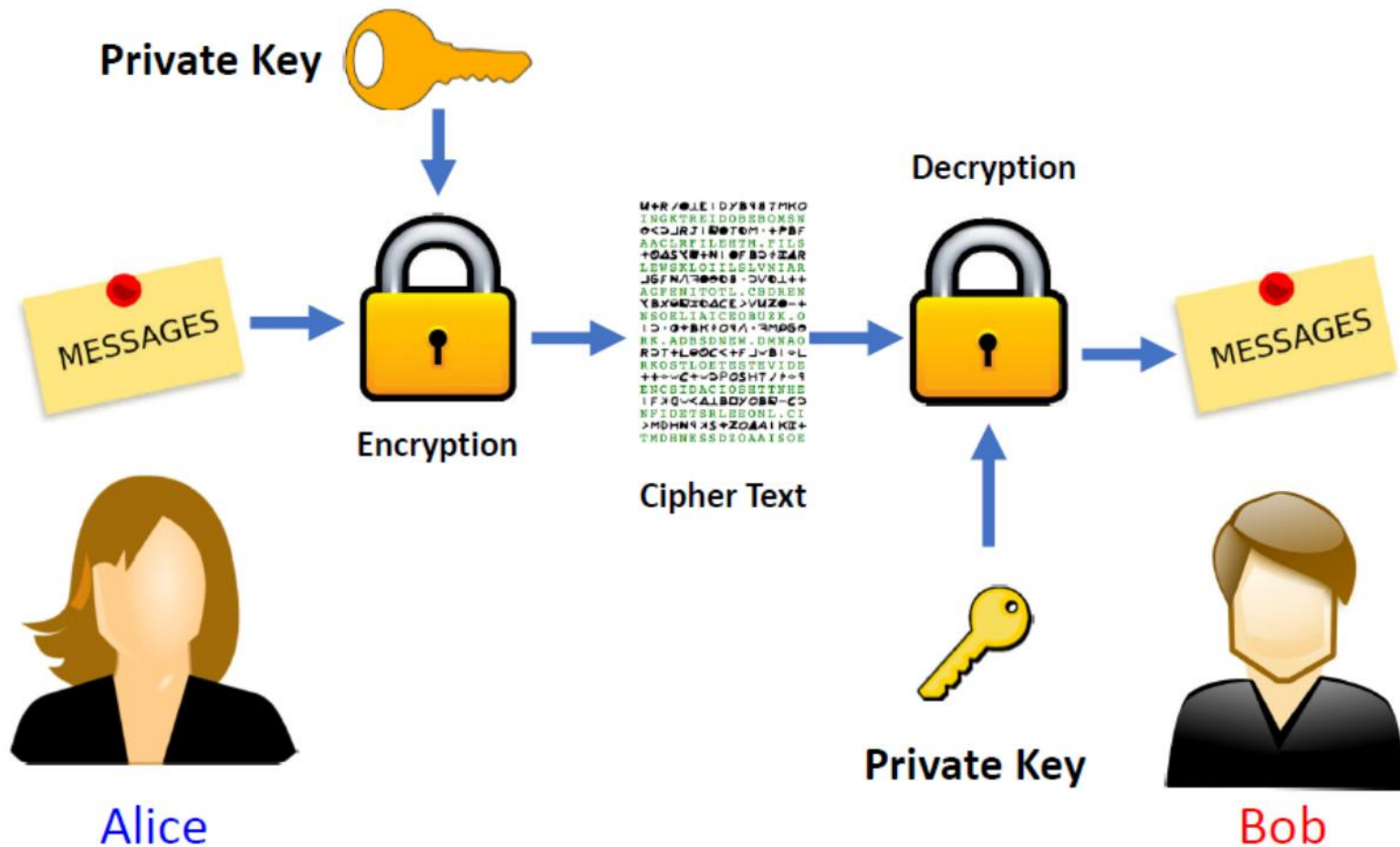


Laplace-Transform
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

- A **cryptogram** is a message written according to a secret code (the Greek word *kryptos* means “hidden”).
- This section describes a method of using matrix multiplication to **encode** and **decode** messages.

Symmetric Cryptography

Laplace-Trans
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$



Cryptography (cont.)



Laplace-Trans
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

- Begin by assigning a number to each letter in the alphabet (with 0 assigned to a blank space), as follows.

0 = _	14 = N
1 = A	15 = O
2 = B	16 = P
3 = C	17 = Q
4 = D	18 = R
5 = E	19 = S
6 = F	20 = T
7 = G	21 = U
8 = H	22 = V
9 = I	23 = W
10 = J	24 = X
11 = K	25 = Y
12 = L	26 = Z
13 = M	

- Then the message is converted to numbers and partitioned into **uncoded row matrices**, each having n entries.

Forming Uncoded Row Matrices

A decorative image in the top right corner showing a close-up of a document with the Laplace Transform formula: $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$. The text "Laplace-Trans" is also visible above the formula.
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

Write the uncoded row matrices of size 1×3 for the message MEET ME MONDAY.

Cryptography (cont.)



Laplace-Trans
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

- To **encode** a message, choose an $n \times n$ invertible matrix A and multiply the uncoded row matrices (on the right) by A to obtain **coded row matrices**.

Encoding a Message

A decorative image in the top right corner showing a close-up of a document with the Laplace Transform formula: $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$. The text "Laplace-Trans" is also visible above the formula.
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

Use the matrix

$$A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$$

to encode the message MEET ME MONDAY.

Cryptography (cont.)

A decorative image in the top right corner showing a portion of a document with the Laplace Transform formula: $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$.

- For those who do not know the matrix A , decoding the cryptogram found in the previous example is difficult.
- But for an authorized receiver who knows the matrix A , decoding is simple.
- The receiver need only multiply the coded row matrices by A^{-1} to retrieve the uncoded row matrices.
- In other words, if

$$X = [x_1 \ x_2 \ \cdots \ x_n]$$

is an uncoded $1 \times n$ matrix, then $Y = XA$ is the corresponding encoded matrix.

- The receiver of the encoded matrix can decode Y by multiplying on the right by A^{-1} to obtain

$$YA^{-1} = (XA)A^{-1} = X.$$

Decoding a Message



Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

Use the inverse of the matrix

$$A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$$

to decode the cryptogram

13 -26 21 33 -53 -12 18 -23 -42 5 -20 56 -24 23 77.

Decoding a Message (cont.)

Q & A

A close-up, slightly blurred image of a document page. The text 'Laplace-Trans' is visible at the top. Below it, the Laplace transform formula is written:
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

with the text 'with the condition' partially visible below the formula.